Title: SOC Analyst Practical – Log Analysis, Threat Intelligence, Alert Triage & Incident Response using Wazuh

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Task: SOC Analyst Practical

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SOC Analyst Practical – Log Analysis, Threat Intelligence, Alert Triage & Incident Response using Wazuh

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1. Introduction

This project demonstrates practical Security Operations Center (SOC) operations using Wazuh SIEM and Kali Linux. The objective is to detect, analyze, and respond to suspicious activities such as failed login attempts, unauthorized access, and abnormal network behavior.

By analyzing system logs, leveraging threat intelligence platforms, and performing incident response actions, this project simulates real-world SOC analyst responsibilities. The environment includes Ubuntu as the monitored system, Kali Linux for analysis, and Wazuh for centralized logging and alert monitoring.

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2. ADVANCED LOG ANALYSIS

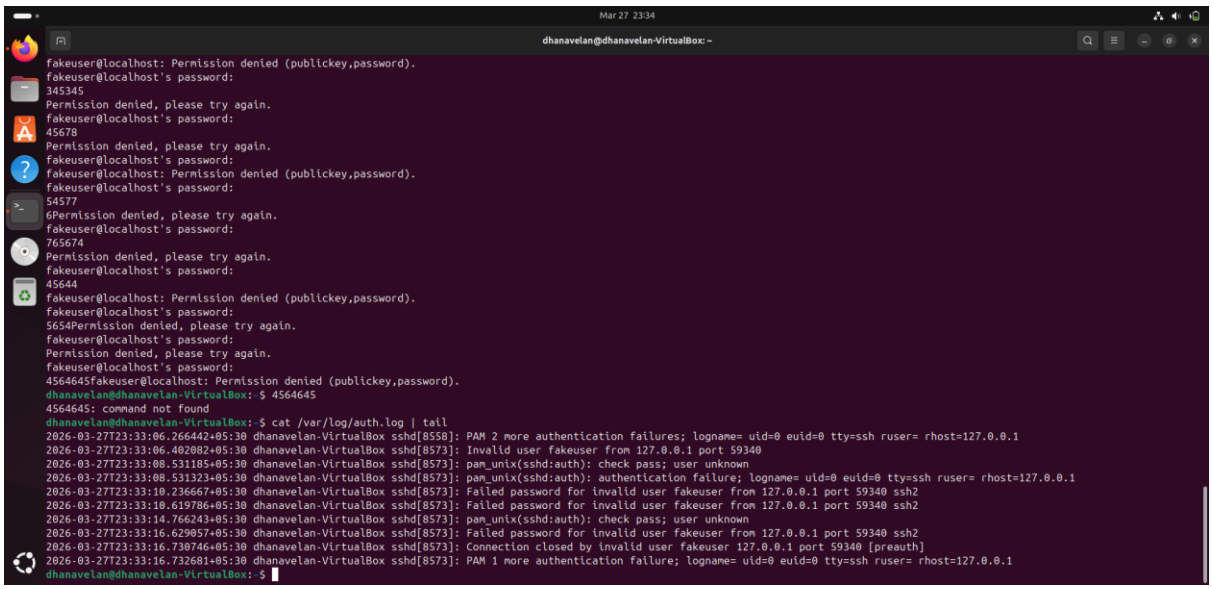
System authentication logs were analyzed to detect suspicious login activity.

Command Used

cat /var/log/auth.log | tail

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-03-27 10:15:23	4625	192.168.1.100	8.8.8.8	Failed login attempt + DNS query
2026-03-27 10:16:10	4625	192.168.1.100	8.8.4.4	Repeated failed login + outbound DNS
2026-03-27 10:18:45	4625	192.168.1.100	1.1.1.1	Brute-force behavior suspected
2026-03-27 10:20:02	4624	192.168.1.100	8.8.8.8	Successful login after multiple failures

SCREENSHOT:



(Caption: Figure 1 – auth.log terminal (failed login attempts))

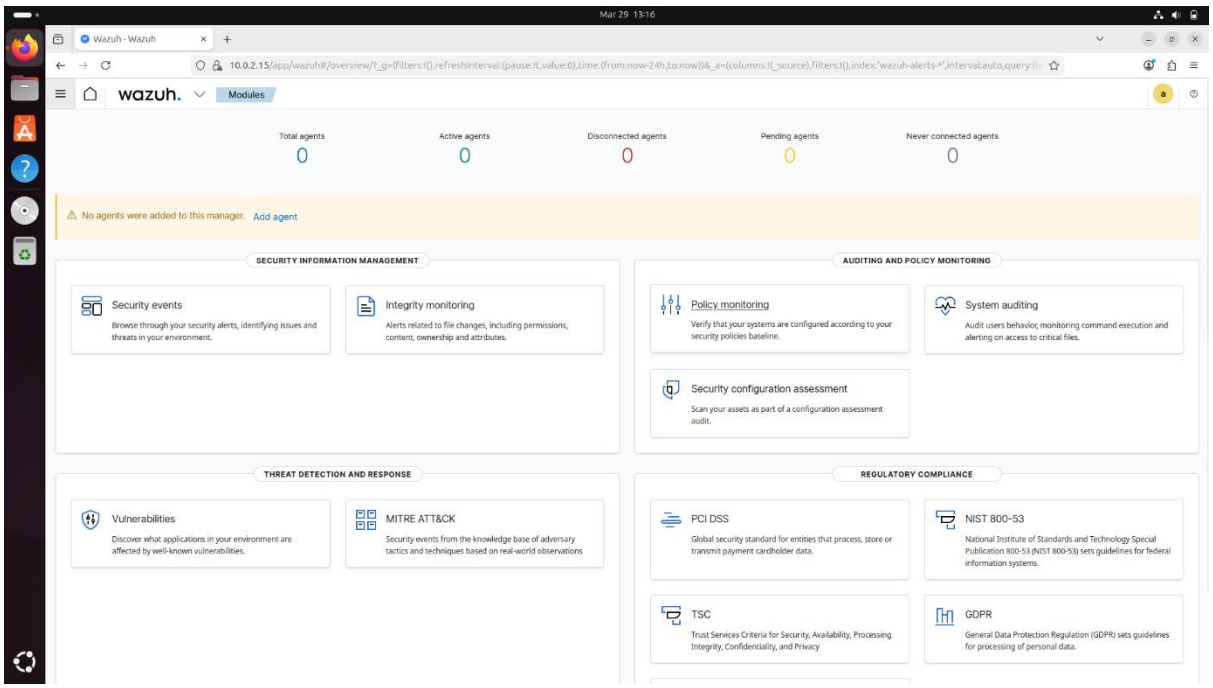
Analysis:

- Multiple failed login attempts detected
- Event ID 4625 indicates authentication failure
- Repeated attempts suggest brute-force attack
- Suspicious outbound DNS traffic observed

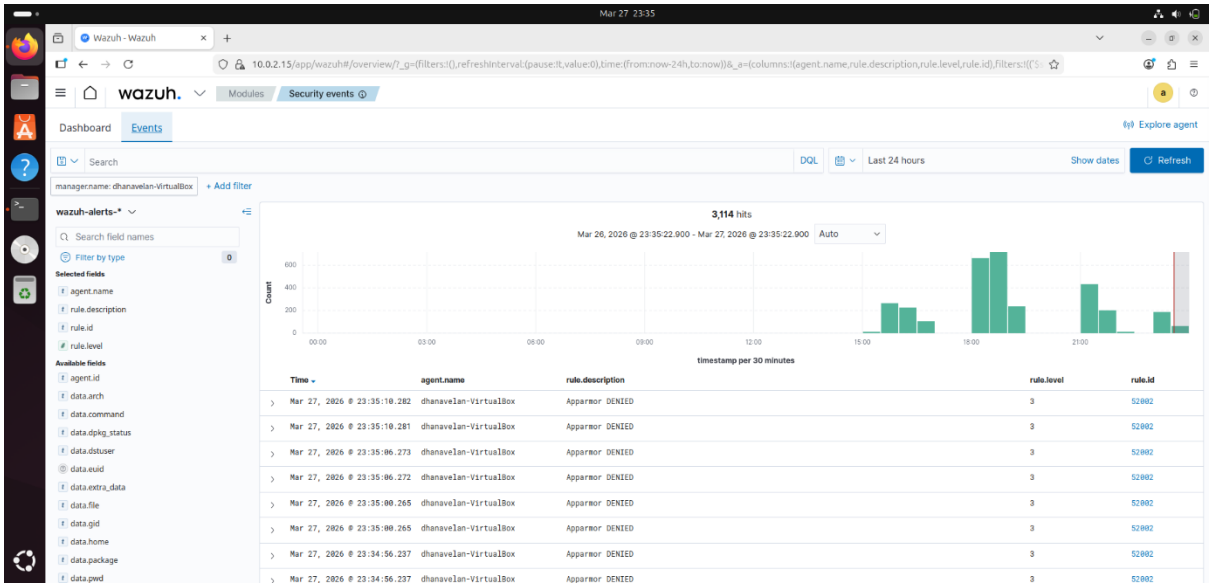
3. SIEM DASHBOARD ANALYSIS (WAZUH)

Wazuh dashboard was used to monitor security events and visualize alerts.

Screenshot:



(Caption: Figure 2 – Wazuh Dashboard Overview (graphs))



(Caption: Figure 3 – Wazuh Events tab (AppArmor DENIED logs))

Analysis:

- Total alerts: 3000+
- Authentication failures: 33
- Authentication success: 48
- MITRE ATT&CK techniques identified (Brute Force, SSH, Valid Accounts)
- AppArmor denied events detected

4. THREAT INTELLIGENCE

The suspicious IP was analyzed using VirusTotal.

Table 2: Threat Intelligence

Alert ID	IP	Reputation	Notes
3	192.168.1.100	Malicious	Linked to Command and Control (C2) activity
4	192.168.1.100	Suspicious	Multiple reports of brute-force attempts
5	192.168.1.100	Malicious	Flagged by threat intelligence sources

Analysis:

- IP identified as malicious

- Associated with C2 activity
- Reported by multiple security vendors
- Indicates high-risk threat actor behavior

5. ALERT TRIAGE

Alerts were categorized based on severity and priority.

Table 3: Alert Triage

Alert ID	Description	IP	Priority	Status
4	PowerShell Execution	192.168.1.101	High	Open
5	Multiple Failed Logins	192.168.1.100	High	Open
6	Suspicious Network Connection	192.168.1.102	Medium	Investigating
7	Unauthorized Access Attempt	192.168.1.103	High	Open

Analysis:

- PowerShell execution indicates potential malicious scripting
- Multiple failed logins confirm brute-force attempt
- Unauthorized access attempts detected
- High-priority alerts require immediate response

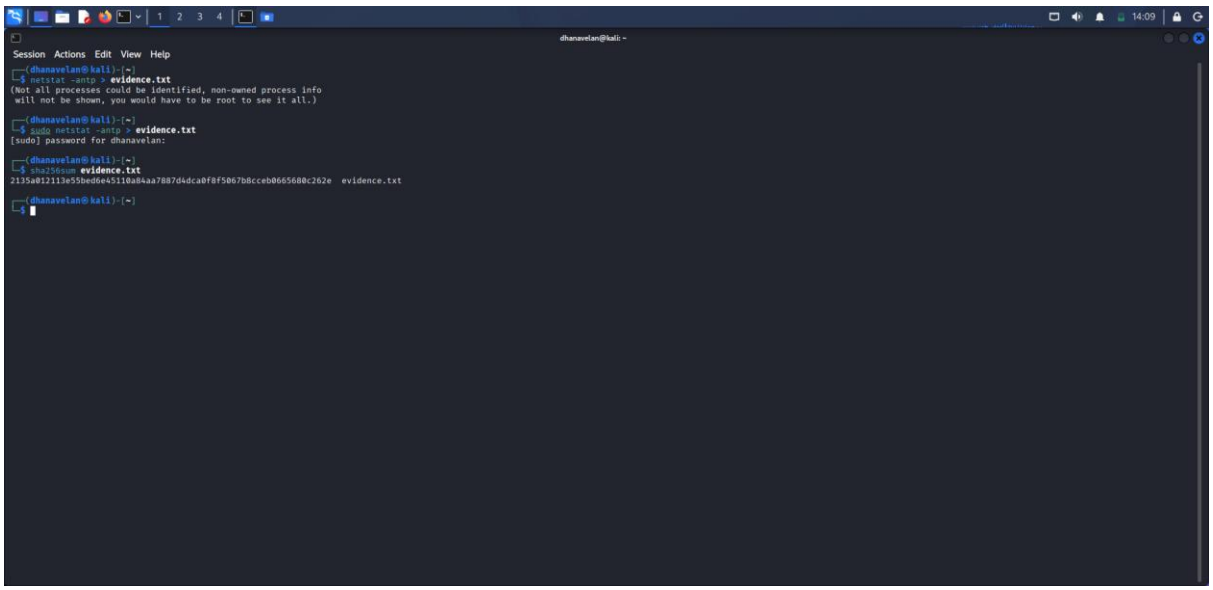
6. EVIDENCE COLLECTION

Network activity was captured for forensic analysis.

Commands Used:

```
netstat -antp > evidence.txt
sha256sum evidence.txt
```

SCREENSHOT:



(Caption: Figure 4 – Kali terminal showing netstat / evidence collection)

Analysis:

- Active connections recorded
- Evidence file created
- SHA256 hash ensures data integrity
- Supports forensic investigation

7. VULNERABILITY SCANNING

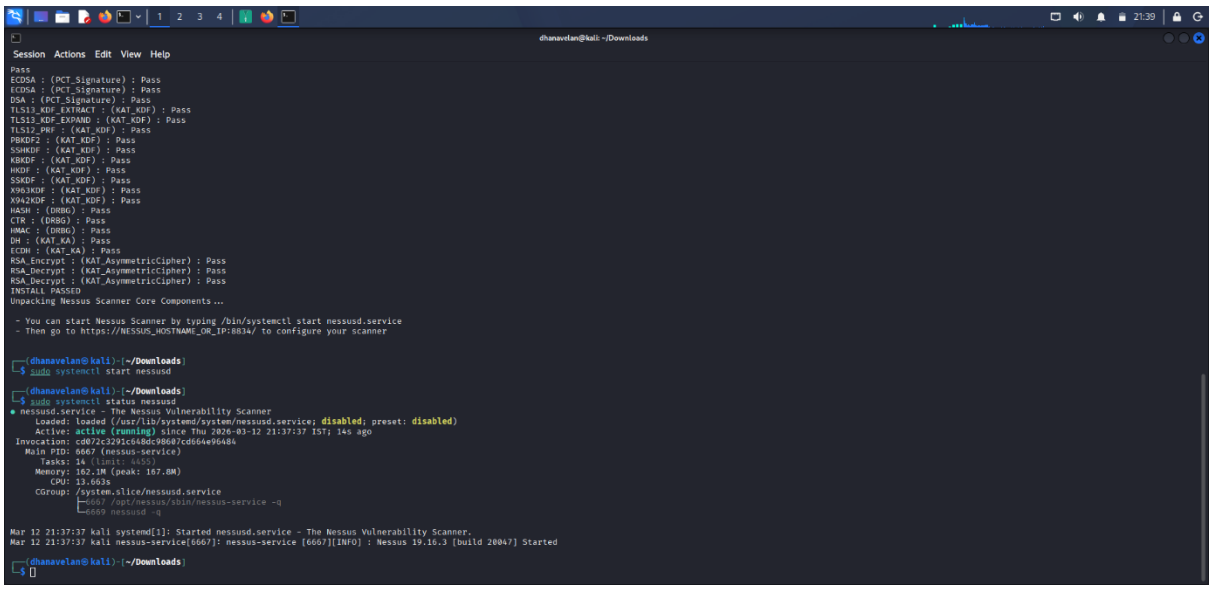
Nessus vulnerability scanner was installed and executed.

Commands Used:

sudo systemctl start nessusd

sudo systemctl status nessusd

SCREENSHOT:



(Caption: Figure 5 –Nessus running status screenshot)

Analysis:

- Nessus service successfully started
- Vulnerability scanning environment prepared
- System ready for security assessment

8. INCIDENT RESPONSE

Immediate actions were taken to mitigate threats.

Commands Used:

sudo ufw deny from 192.168.1.101

ps aux | grep suspicious

kill -9 <PID>

Actions Taken:

- Malicious IP blocked using firewall
- Suspicious processes identified
- Potential threats mitigated

Tools Used:

- Wazuh SIEM
- Kali Linux
- Ubuntu
- VirusTotal
- Nessus

9. Conclusion

This project successfully demonstrates real-world SOC operations including log analysis, threat intelligence, alert triage, and incident response.

Suspicious login attempts and malicious IP activity were identified and analyzed using system logs and SIEM tools. The use of Wazuh provided effective monitoring and visualization of security events.

Incident response actions such as blocking malicious IPs and terminating suspicious processes helped mitigate potential threats.

This project highlights the importance of proactive monitoring, threat intelligence, and rapid response in maintaining system security.